Exercise 10.4 - AI4I Data Analysis

Introduction

In this report, the AI4I 2020 Predictive Maintenance Dataset is analyzed using Pandas, Matplotlib and Seaborn.

The dataset contains 10,000 measurements from a simulated industrial machine.

Data Preparation

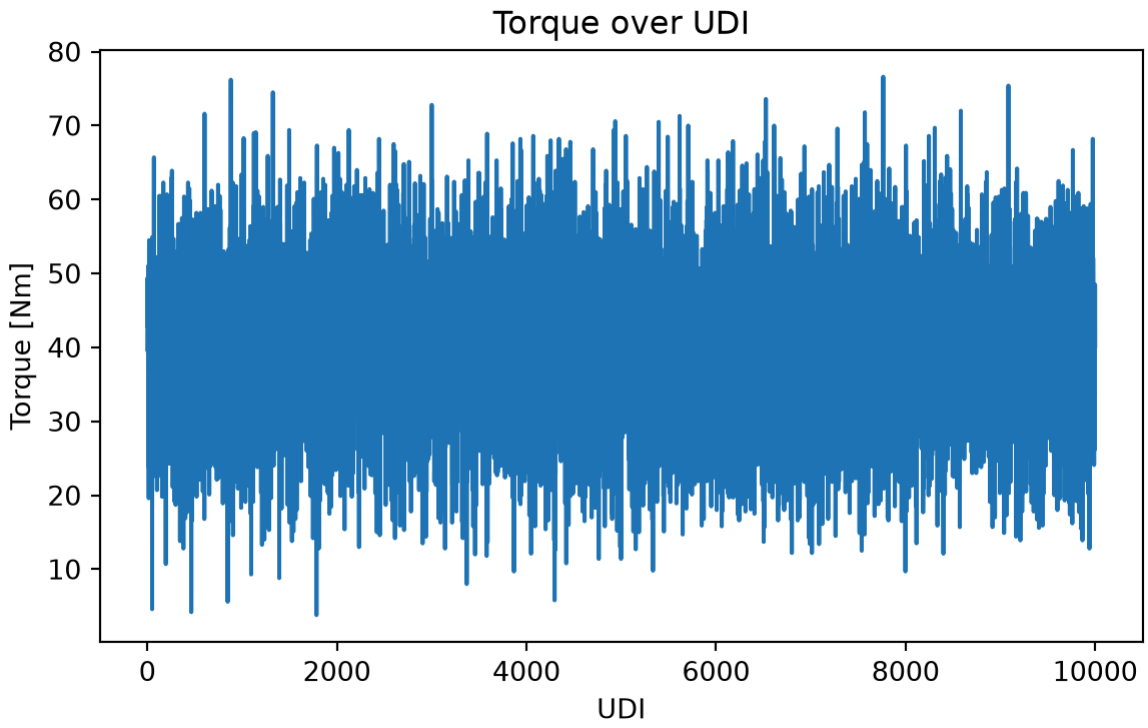
▶ Show Python code

	UDI	Torque [Nm]	Rotational speed [rpm]	Air temperature [K]
0	1	42.8	1551	298.1
1	2	46.3	1408	298.2
2	3	49.4	1498	298.1
3	4	39.5	1433	298.2
4	5	40.0	1408	298.2

Basic Plots

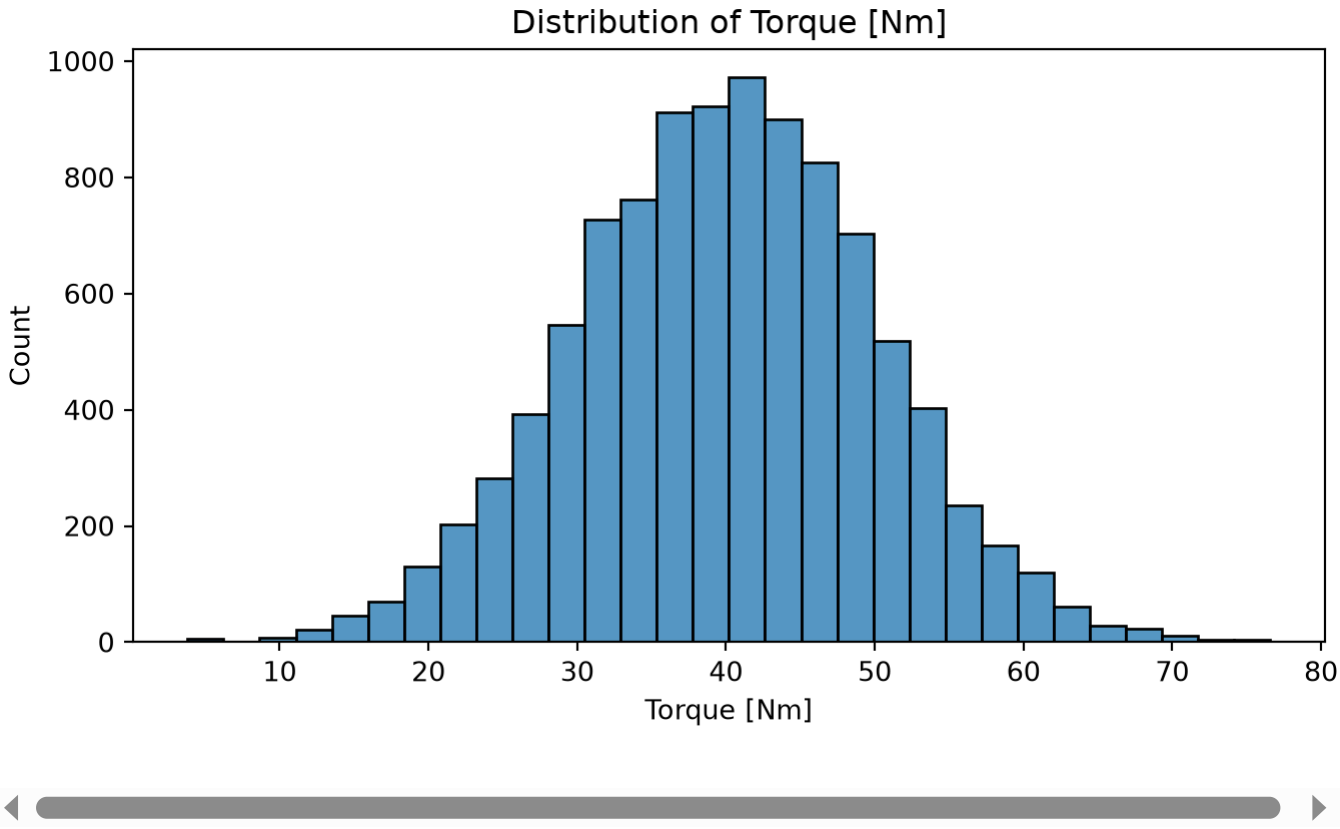
Lineplot

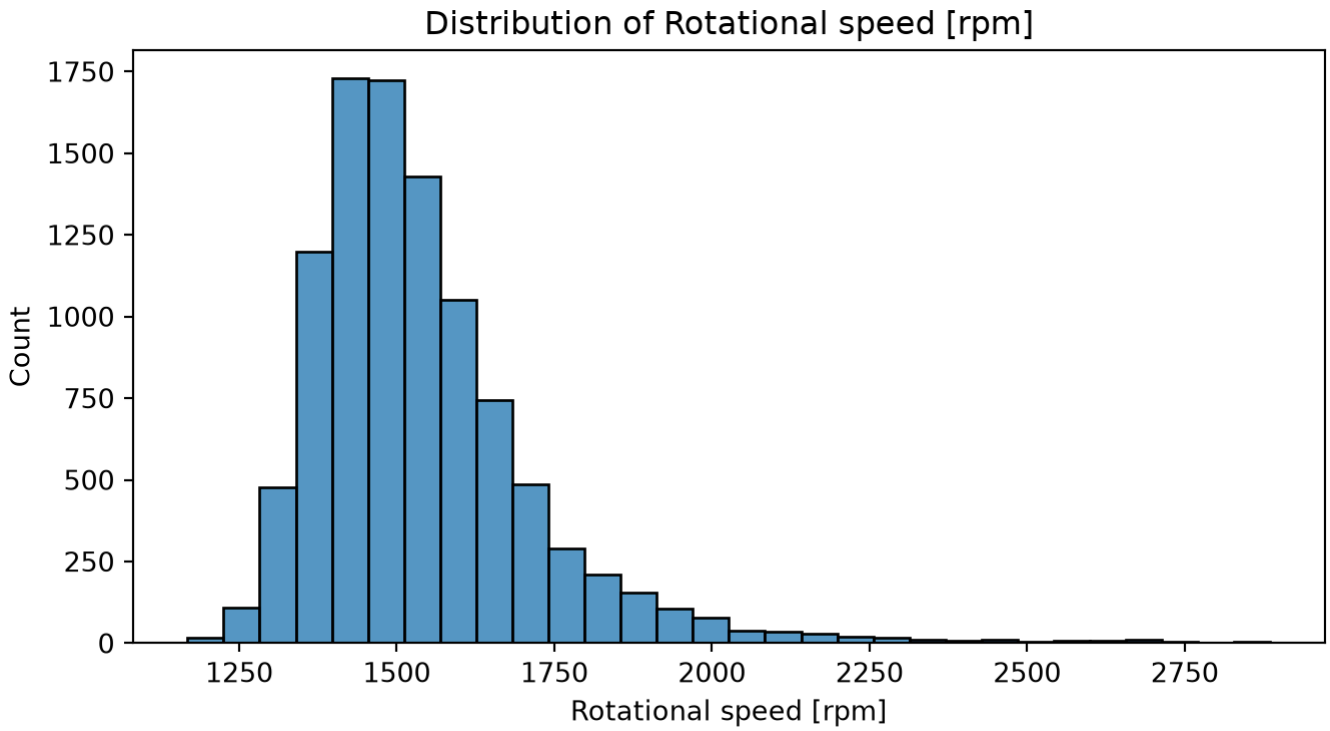
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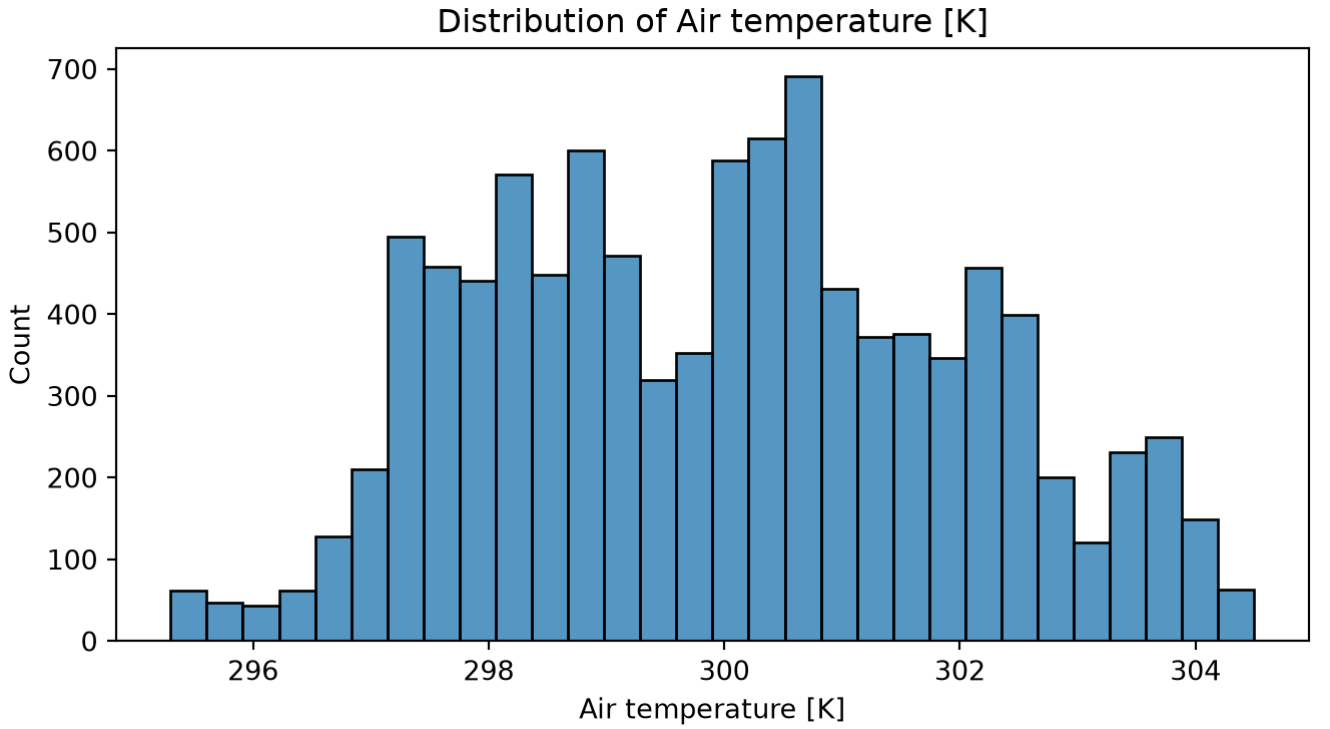


Histograms

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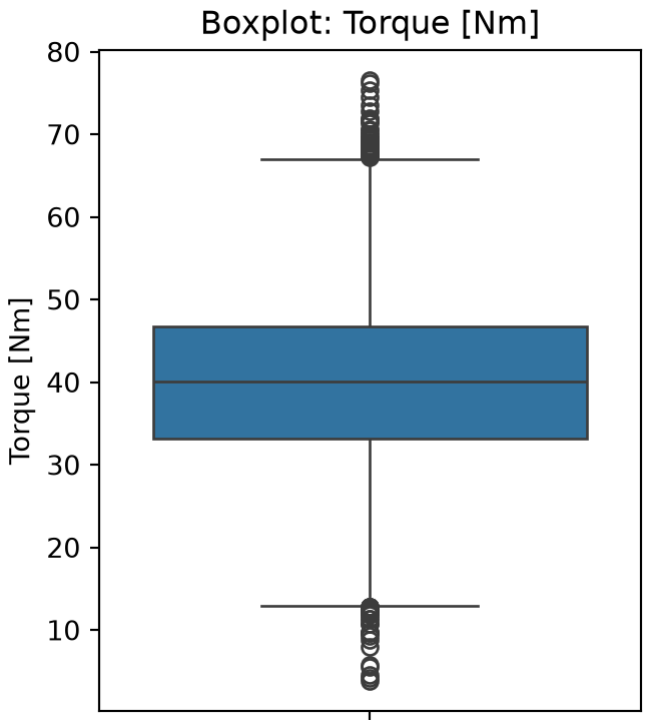


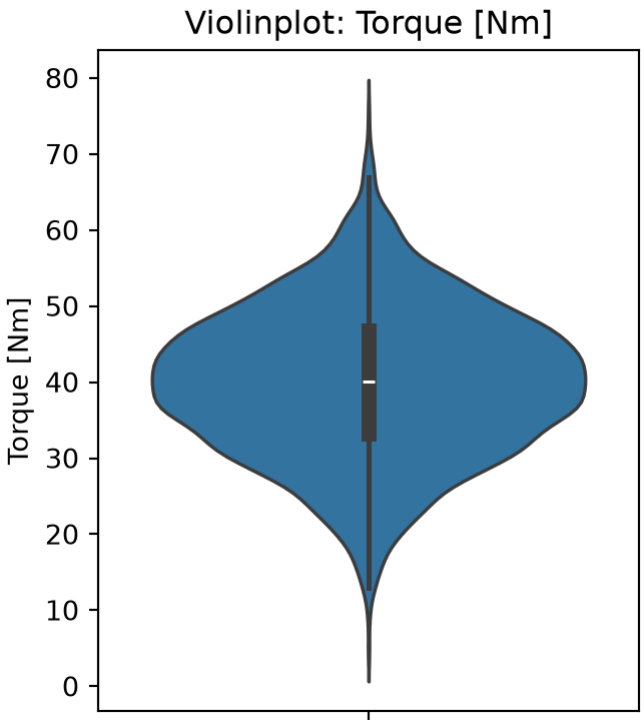


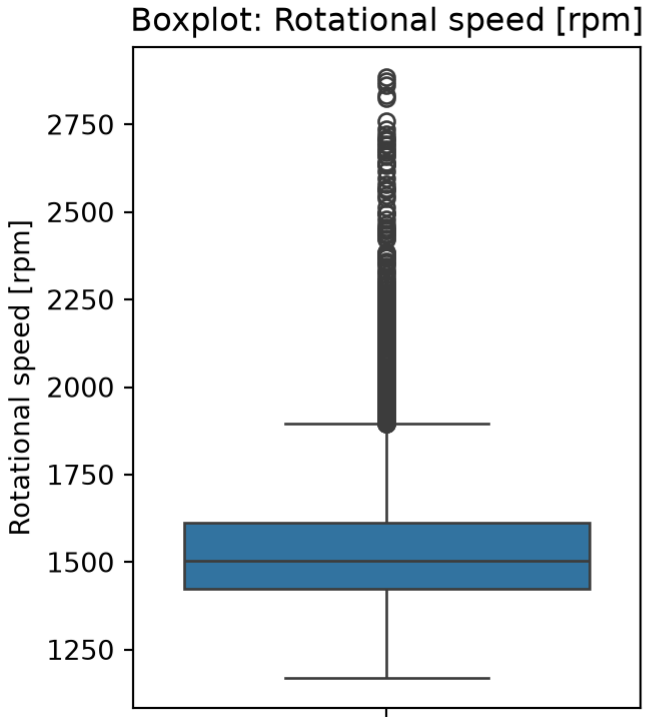
Boxplots and Violinplots

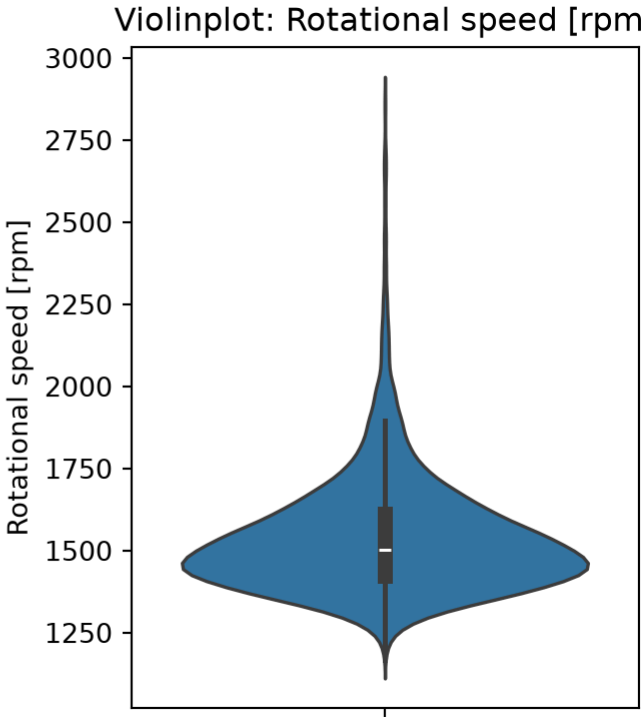
The boxplot summarizes statistical values, while the violinplot also shows the estimated distribution of measurements.

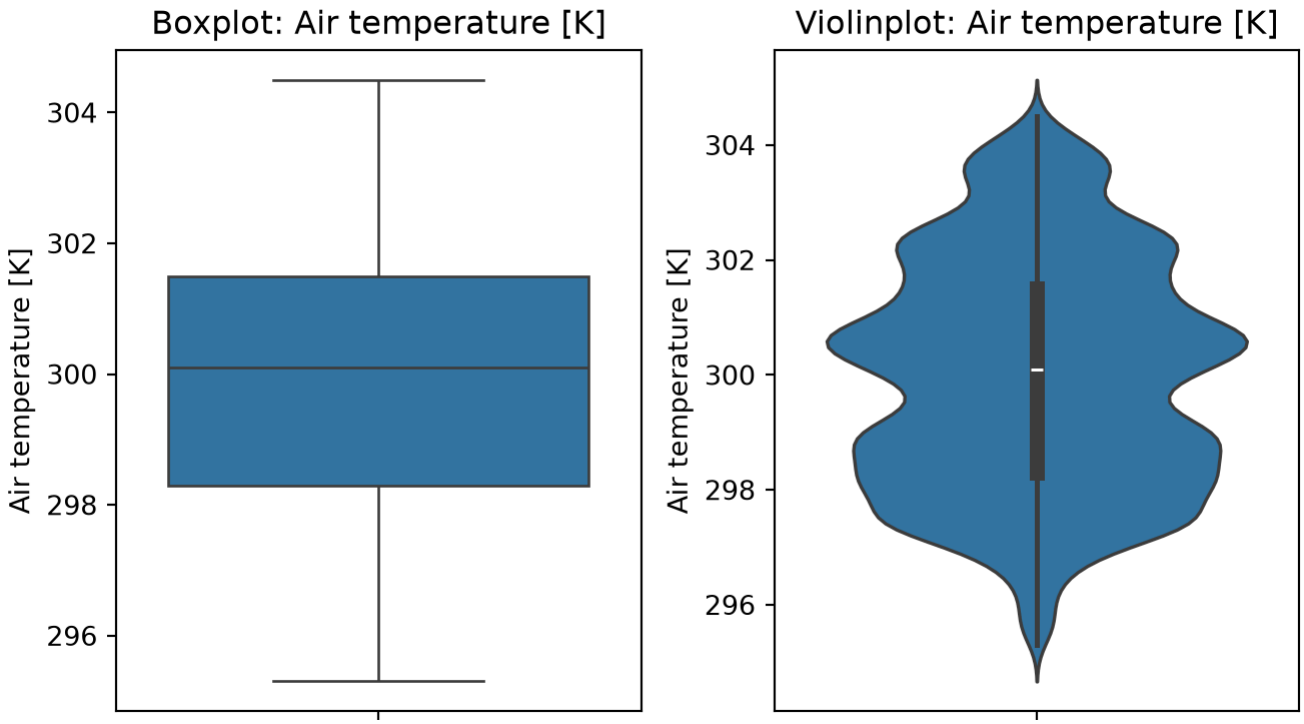
▶ Show Python code


Boxplot: Torque [Nm]
Torque [Nm]
80
70
60
50
40
30
20
10


Violinplot: Torque [Nm]
Torque [Nm]
80
70
60
50
40
30
20
10
0


Boxplot: Rotational speed [rpm]
Rotational speed [rpm]
2750
2500
2250
2000
1750
1500
1250

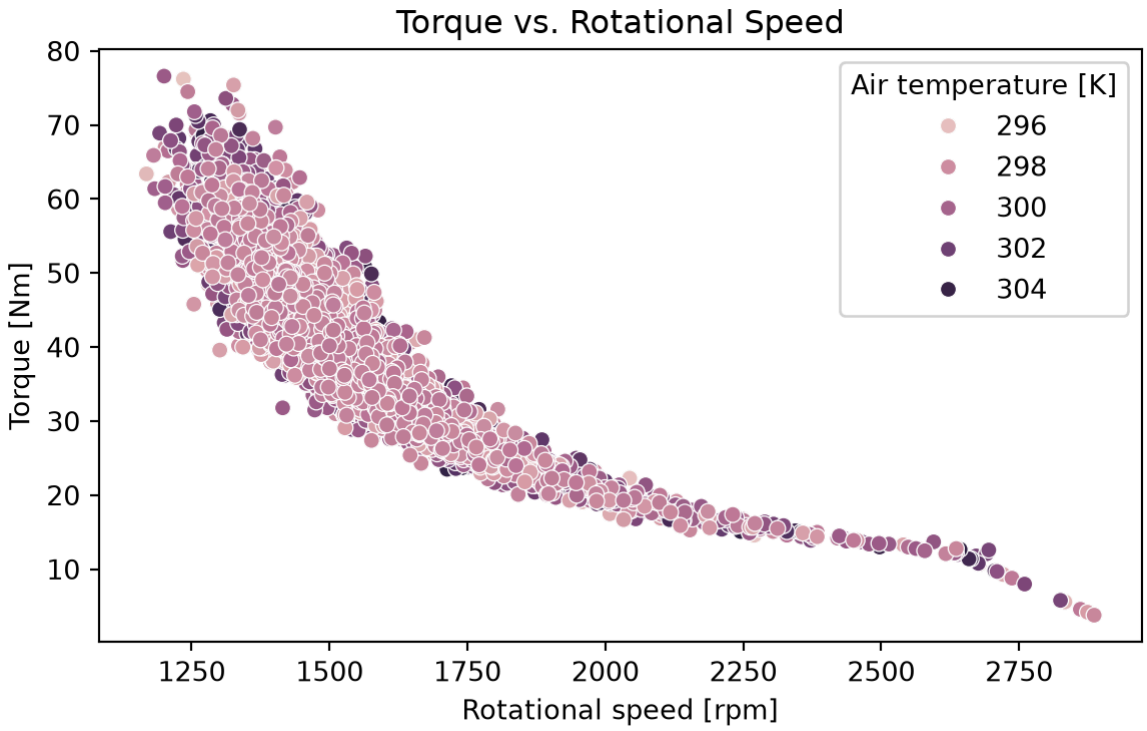

Violinplot: Rotational speed [rpm]
Rotational speed [rpm]
3000
2750
2500
2250
2000
1750
1500
1250



Relationships Between Variables

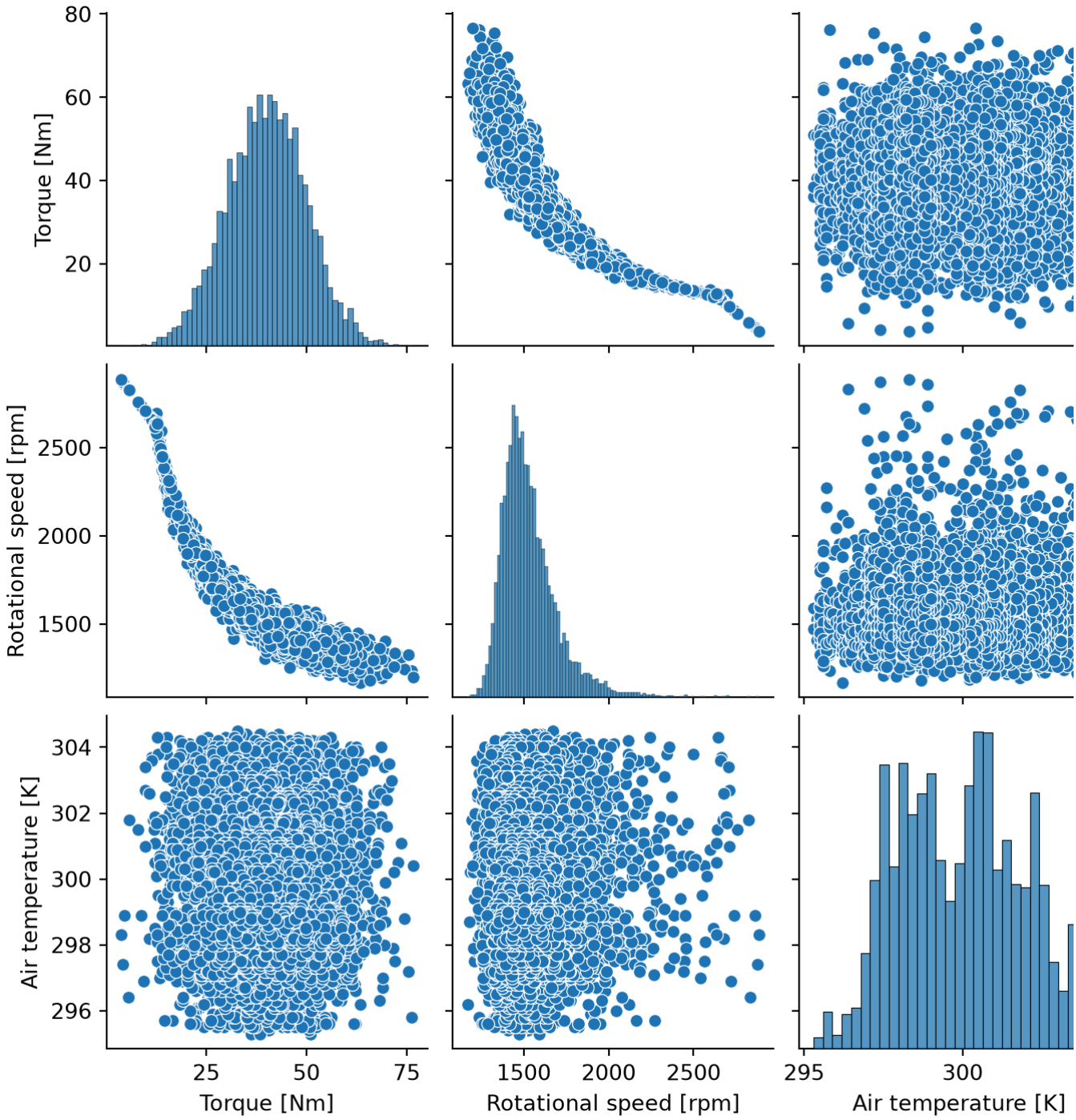
Scatterplot

▶ Show Python code



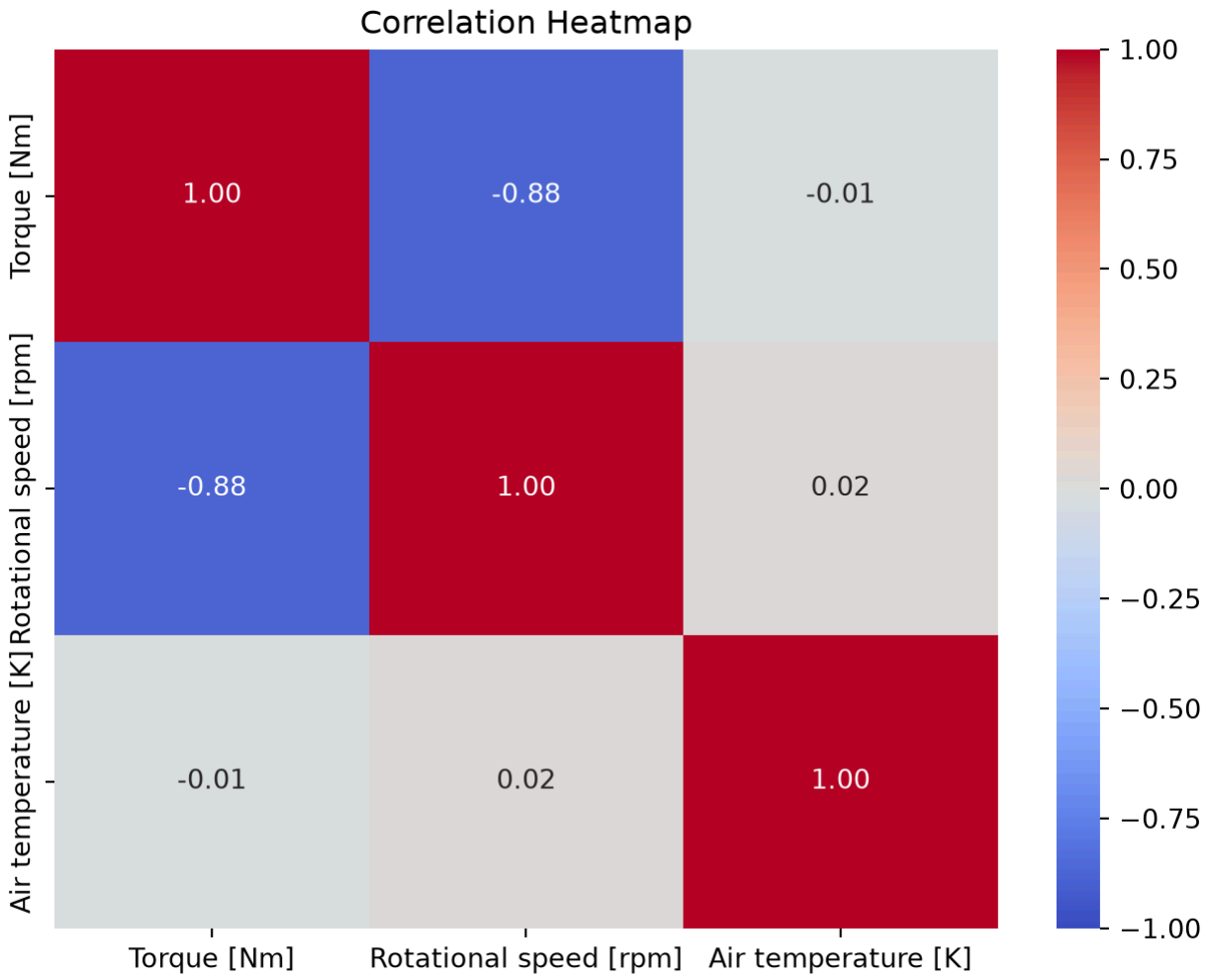
Pairplot

▶ Show Python code



Correlation Heatmap

▶ Show Python code

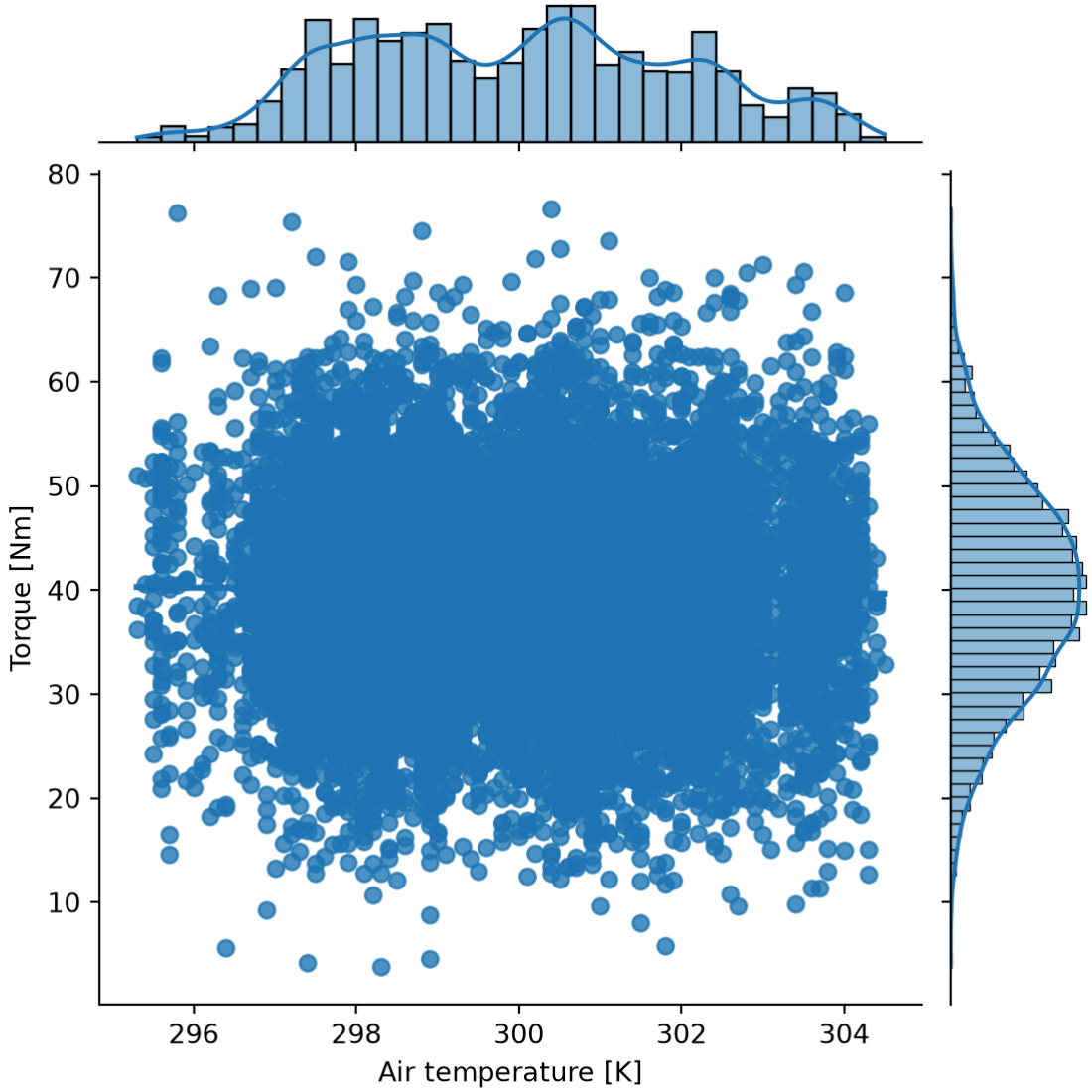


The Pearson correlation coefficient ranges from -1 to +1. Negative values indicate a negative linear relationship, while positive values indicate a positive linear relationship.

Advanced Plots

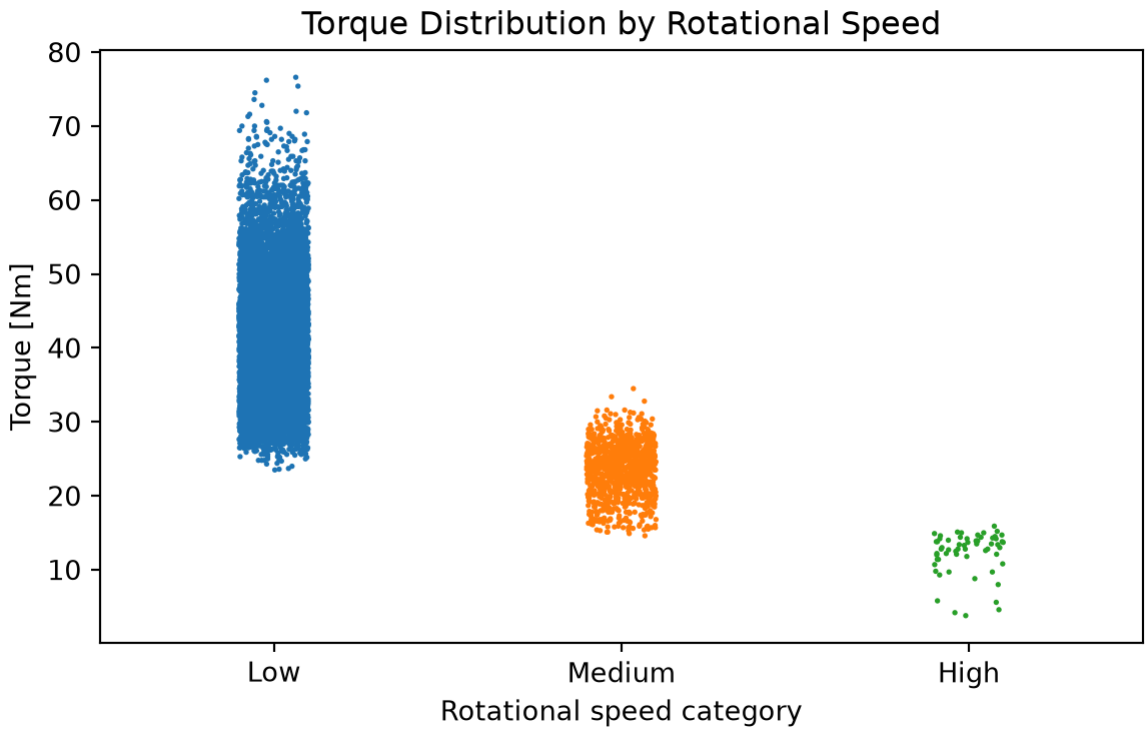
Jointplot

▶ Show Python code



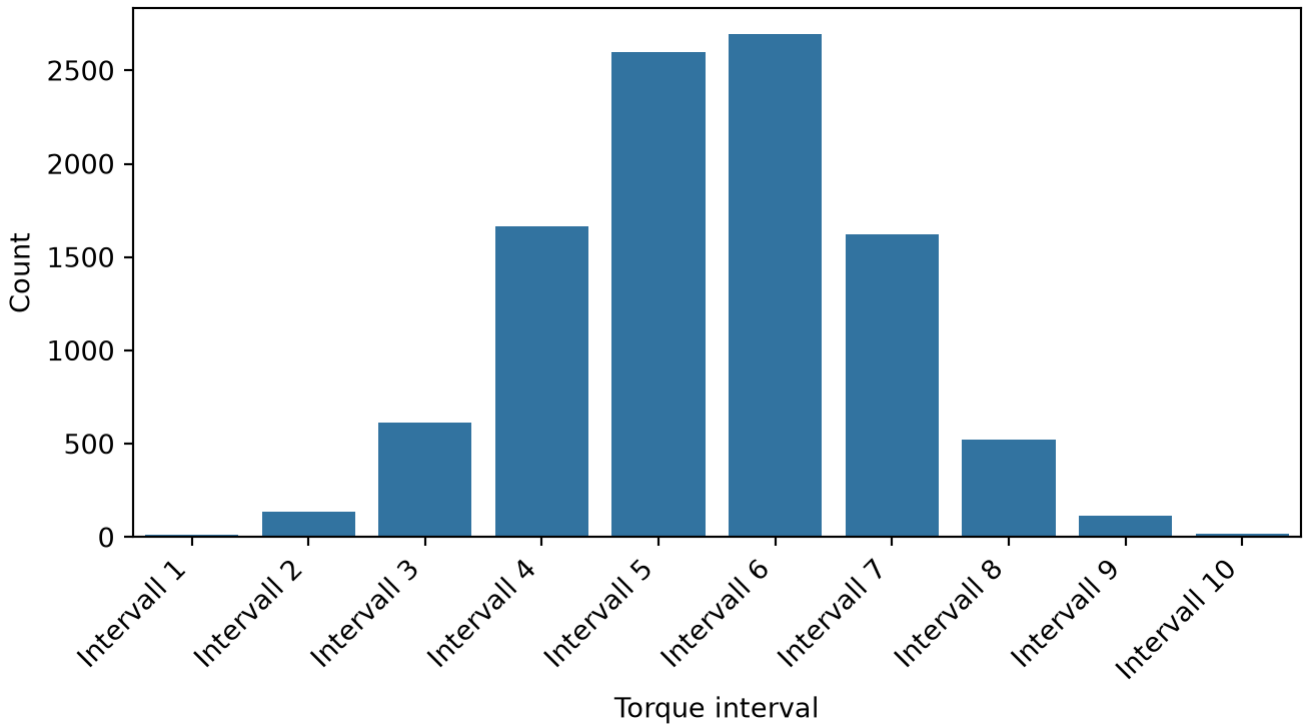
Stripplot

▶ Show Python code



Countplot

▶ Show Python code



Observations

Are there any obvious outliers in the data?

Yes, the boxplots show that especially torque and rotational speed have some outliers.

Which two variables seem most related?

Torque and rotational speed show a negative relationship.

Are there any non-linear patterns visible?

No clear non-linear patterns were identified in the scatterplots.

How are the distributions shaped?

Torque is approximately normally distributed, with most measurements around 40 Nm.

Rotational speed has a right-skewed distribution, with most values around 1500 rpm.

Air temperature does not show a clear normal distribution and has an irregular distribution over the measured range.